MCO — METACHRONAL CASCADE ORNITHOPTER

Wing Manufacturing Document

MCO-1D Drone Prototype

S1223 Airfoil Specification · Universal LW-PLA Rib Design · SolidWorks Parameters · Bill of Materials · Assembly Instructions

Design philosophy	One universal rib design at 120mm chord. Membrane trimmed to each wing's planform. Ribs left full-length as internal formers. Aerodynamic shape defined by the membrane, not the rib edge.
Performance	Approximately 1-2% efficiency reduction versus the fully optimised 12-configuration design. Acceptable for prototype. Recoverable on second build.
Build count	110 identical ribs across all 8 wings. One slicer setup. One print run of approximately 15-22 hours. No trimming of ribs required.
Document structure	Chapter 1: Airfoil specification and properties. Chapter 2: SolidWorks rib parameters and feature tree. Chapter 3: Coordinate file for profile import. Chapter 4: Bill of materials. Chapter 5: Assembly instructions step by step. Blueprint: Annotated cross-section schematic.

CHAPTER 1 — THE S1223 AIRFOIL — SPECIFICATION AND PROPERTIES

1.1 Why S1223

The Selig S1223 was designed specifically for high-lift low-Reynolds-number applications. It is the correct choice for the MCO-1D drone operating at chord Reynolds numbers of 3,500 to 26,000 across the cascade. It provides the highest maximum lift coefficient of any section in its class while retaining a soft stall characteristic — stall progresses gradually from root to tip rather than snapping off abruptly. This soft stall behaviour is essential for a flapping wing that operates at high angles of attack on every power stroke.

1.2 Key Geometric Properties

Designation	Selig S1223
Source	UIUC Airfoil Database — s1223.dat
Maximum thickness	12.14% chord at 30.08% chord position
Maximum camber	8.68% chord at 47.32% chord position
Leading edge radius	1.56% chord
Trailing edge	Blunted at 97% chord — 0.80mm flat close (printability)
At 120mm chord:	
Max thickness	14.57mm at 36.1mm from leading edge
Max camber	10.42mm at 56.8mm from leading edge
LE radius	1.87mm radius
TE close point	116.4mm from LE (97% chord)
TE close width	0.80mm flat vertical close
Design Reynolds number	100,000 to 300,000 (original design range)

MCO operating Re	3,500 to 26,600 — below design range; S1223 still outperforms alternatives at these Re due to high camber maintaining attached flow with LEV assistance
CL max (experimental)	2.4 to 2.6 at Re 100,000 (with VGs and covert flaps)
Best L/D AoA	8 to 10 degrees — used for glide and WING-RANGE modes
Power stroke AoA	35 to 42 degrees — deep post-stall, LEV-dominated
Stall character	Soft stall — root first, gradual progression outboard

1.3 The Trailing Edge Modification

The original S1223 has a trailing edge of approximately 0.25% chord thickness at the tip — 0.30mm at 120mm chord. This cannot be printed on a 0.4mm nozzle. The modification truncates the profile at 97% chord (116.4mm from the LE) and closes with a vertical 0.80mm flat face. This is standard practice in model aircraft construction and has been shown to have negligible aerodynamic effect below Reynolds number 200,000. The blunt TE also provides a clean, consistent attachment point for the trailing edge carbon strip and membrane.

The 97% chord truncation is already incorporated into all coordinate tables in this document. No further modification is required in SolidWorks. The coordinate file terminates at x = 116.4mm, not 120mm.

CHAPTER 2 — SOLIDWORKS RIB PARAMETERS

2.1 Global Variables

Enter these in SolidWorks via Tools → Equations → Global Variables. Every feature dimension references these variables. The rib is fully defined by these values with no additional inputs.

Variable	Value	Unit	Purpose
Chord_mm	120.00	mm	Master chord length. Do not change for the universal rib.
Rib_thickness	2.00	mm	Extrusion depth. Constant for all ribs.
LE_bore_dia	2.10	mm	LE spar clearance bore. Fits 2.0mm OD spar.
Sec_spar_dia	3.10	mm	Secondary spar notch diameter. Fits 2.0mm OD rod.
Sec_spar_x	72.00	mm	Secondary spar X position (60% of 120mm chord).
TE_x	116.40	mm	Trailing edge truncation point (97% chord).
TE_close	0.80	mm	Blunt TE close width.
TE_chamfer_run	0.40	mm	TE chamfer distance.
TE_chamfer_angle	30.00	deg	TE chamfer angle.
LE_solid_x	18.00	mm	LE solid zone boundary (15% of 120mm).
Mid_zone_end_x	96.00	mm	Mid zone end (80% of 120mm).
Flap_row1_x	45.00	mm	Film flap row 1 forward edge (37.5% chord).
Flap_row2_x	63.00	mm	Film flap row 2 forward edge (52.5% chord).
Flap_row3_x	81.00	mm	Film flap row 3 forward edge (67.5% chord).
Flap_width	9.60	mm	Film flap chord width (8% of 120mm).
Flap_recess_depth	0.30	mm	Film flap bond ledge recess depth.
VG_x	20.40	mm	VG recess centre X (17% of 120mm chord).
VG_recess_depth	0.20	mm	VG recess depth.
VG_recess_width	2.00	mm	VG recess chord width.

Variable	Value	Unit	Purpose
VG_dimple_dia	0.30	mm	VG alignment dimple diameter.
VG_dimple_depth	0.15	mm	VG alignment dimple depth.
Bore_fillet_r	0.30	mm	Fillet radius on LE bore edges.

2.2 Feature Tree — Step by Step

Build features in this exact order. Each step references the global variables defined above.

STEP 1 — Create New Part

File → New → Part. Set document units to MILLIMETRES (Tools → Options → Document Properties → Units). Enter all Global Variables from Section 2.1.

STEP 2 — Import Profile (Base Sketch)

Insert → Curve → Curve Through XYZ Points. Browse to the coordinate file from Chapter 3. SolidWorks imports 61 points and fits a spline. The result is a closed profile on the Front Plane with LE at the origin.

STEP 3 — Base Extrusion

Select the closed profile. Features → Extruded Boss/Base. Depth = 'Rib_thickness' = 2.00mm. Direction: along the Z axis. Result: 2.00mm thick S1223 solid.

STEP 4 — LE Spar Bore

Sketch2 on front face of rib. Draw circle: diameter = 'LE_bore_dia' = 2.10mm. Centre: X = 1.80mm, Y = 1.85mm (above chord line). Features → Extruded Cut → Through All Both Directions. Then add Fillet: radius = 'Bore_fillet_r' = 0.30mm on both bore entry edges.

STEP 5 — Secondary Spar Notch

Sketch3 on TOP face (upper surface). Draw semicircle: diameter = 'Sec_spar_dia' = 3.10mm, open end facing upward. Centre X = 'Sec_spar_x' = 72.00mm. Centre Y: use Pierce constraint to the upper surface spline (approximately 11.60mm above chord line at 60% chord). Cut: 2.00mm deep downward through rib.

STEP 6 — Trailing Edge Chamfer

Select both TE edges (upper and lower faces at x = 116.4mm). Insert → Features → Chamfer. Distance = 'TE_chamfer_run' = 0.40mm. Angle = 'TE_chamfer_angle' = 30°. Apply to both edges.

STEP 7 — Film Flap Ledge Row 1

Sketch4 on upper face. Rectangle: chord-width = 'Flap_width' = 9.60mm, span-width = 'Rib_thickness' = 2.00mm. Forward edge at X = 'Flap_row1_x' = 45.00mm. Use Pierce constraint to position Y on upper surface spline. Cut: 'Flap_recess_depth' = 0.30mm deep perpendicular to upper surface.

STEP 8 — Film Flap Ledge Row 2

Identical to Step 7. Forward edge at X = 'Flap_row2_x' = 63.00mm.

STEP 9 — Film Flap Ledge Row 3

Identical to Steps 7-8. Forward edge at X = 'Flap_row3_x' = 81.00mm.

STEP 10 — VG Recess

Sketch7 on upper face. Rectangle: chord-width = 'VG_recess_width' = 2.00mm, span-width = 2.00mm. Centre at X = 'VG_x' = 20.40mm. Pierce to upper surface spline for Y. Cut: 'VG_recess_depth' = 0.20mm deep.

STEP 11 — VG Alignment Dimple

Sketch8 on the floor of the VG recess. Draw point at centre. Insert → Features → Revolved Cut. Profile: right triangle with base 0.15mm, height 0.15mm, 60° included angle. Revolve 360°. Result: conical dimple 0.30mm dia × 0.15mm deep at VG recess centre.

STEP 12 — Split Bodies for Infill Zones

Insert → Features → Split. Trim tool 1: Reference Plane offset from Front Plane at X = 'LE_solid_x' = 18.00mm (15% chord). Trim tool 2: Reference Plane at X = 'Mid_zone_end_x' = 96.00mm (80% chord). Result: three separate bodies. Name them: Body1-LE-Solid, Body2-Mid-Gyroid, Body3-TE-Gyroid.

STEP 13 — Save and Export

File → Save As → MCO_UniversalRib_120mm.SLDPRT. Then export as STL: File → Save As → STL. Set resolution to Fine. Export all three bodies as separate STL files: MCO_Rib_Body1_LESolid.stl, MCO_Rib_Body2_MidGyroid.stl, MCO_Rib_Body3_TEGyroid.stl.

2.3 Critical Feature Positions Summary

Feature	Position	Depth / Height	Dimensions
LE spar bore centre	X = 1.80mm, Y = 1.85mm	Through all	2.10mm dia + 0.30mm edge fillet
Secondary spar notch	X = 72.00mm, Y = upper surface	2.00mm deep	3.10mm dia semicircle
LE solid zone boundary	X = 18.00mm (15%c)	Split plane	Infill zone separator
Mid/TE zone boundary	X = 96.00mm (80%c)	Split plane	Infill zone separator
Film flap ledge R1	X = 45.00mm (37.5%c)	0.30mm deep	9.60mm wide × 2.00mm span
Film flap ledge R2	X = 63.00mm (52.5%c)	0.30mm deep	9.60mm wide × 2.00mm span
Film flap ledge R3	X = 81.00mm (67.5%c)	0.30mm deep	9.60mm wide × 2.00mm span
VG recess centre	X = 20.40mm (17%c)	0.20mm deep	2.00mm × 2.00mm + dimple
TE truncation	X = 116.40mm (97%c)	0.80mm close	Blunt TE close
TE chamfer both edges	X = 116.40mm	0.40mm run, 30°	Upper and lower faces

CHAPTER 3 — COORDINATE FILE FOR SOLIDWORKS IMPORT

Save the following as a plain text file named MCO_S1223_120mm.sldcrv (or .txt). Three columns: X, Y, Z. Space-separated. No headers. Import via Insert → Curve → Curve Through XYZ Points. All values in millimetres. Z = 0.000 throughout. 61 points form the complete closed profile including the blunt TE close.

The upper surface runs from the LE (0,0) to the TE (116.4, 0). The TE close line adds one intermediate point at (116.4, -0.8). The lower surface returns from the TE to the LE. SolidWorks fits a single closed spline through all 61 points in order.

UPPER SURFACE — 30 points (LE to TE):

0.000	0.000	0.000
0.151	1.001	0.000
0.542	1.958	0.000
1.522	3.416	0.000
3.048	4.914	0.000
5.729	6.872	0.000
9.253	8.771	0.000
13.516	10.522	0.000
18.388	12.047	0.000
23.738	13.297	0.000
29.436	14.244	0.000
36.124	14.861	0.000
41.326	15.108	0.000
47.244	14.996	0.000
52.963	14.544	0.000
58.354	13.791	0.000
63.293	12.791	0.000
67.629	11.597	0.000
71.337	10.273	0.000

74.141	8.896	0.000
78.918	7.291	0.000
84.101	5.616	0.000
89.597	4.049	0.000
95.174	2.686	0.000
100.533	1.622	0.000
105.640	0.878	0.000
110.054	0.415	0.000
113.975	0.137	0.000
116.622	0.002	0.000
116.400	0.000	0.000

TRAILING EDGE CLOSE — 2 points:

116.400	0.000	0.000
116.400	-0.800	0.000

LOWER SURFACE — 29 points (TE back to LE):

116.400	-0.800	0.000
116.622	-0.036	0.000
113.975	-0.143	0.000
110.054	-0.393	0.000
105.640	-0.623	0.000
100.533	-0.798	0.000
95.174	-0.917	0.000
89.597	-0.979	0.000
84.101	-0.972	0.000
78.918	-0.878	0.000
74.141	-0.769	0.000
71.337	-0.630	0.000

67.629	-0.498	0.000
63.293	-0.357	0.000
58.354	-0.214	0.000
52.963	-0.099	0.000
47.244	-0.009	0.000
41.326	0.040	0.000
36.124	0.076	0.000
29.436	0.075	0.000
23.738	0.042	0.000
18.388	-0.017	0.000
13.516	-0.088	0.000
9.253	-0.194	0.000
5.729	-0.318	0.000
3.048	-0.396	0.000
1.522	-0.339	0.000
0.542	-0.206	0.000
0.151	-0.092	0.000
0.000	0.000	0.000

Once imported the curve should appear as a smooth closed teardrop shape with the LE at the origin, chord running right along the X axis, and the camber curving upward. If the curve appears distorted, check that document units are set to millimetres before importing.

CHAPTER 4 — 3D PRINT SETTINGS

Material	ColorFabb LW-PLA or Spectrum Ultra Foam PLA
Nozzle temperature	252°C (full activation, maximum foaming)
Bed temperature	60°C, smooth PEI sheet
Layer height	0.15mm
Nozzle diameter	0.40mm
Print speed	35 mm/s
Fan speed	80% from layer 3 onward
Extrusion multiplier	0.67 (tune ±0.02 on first test rib)
Print orientation	Rib FLAT on bed: chord horizontal, span (Z axis) vertical. This aligns print layers with aerodynamic load direction.
Supports	NONE required in this orientation
Top / bottom layers	2 solid layers
Body 1 infill	100% solid (LE nose zone 0-18mm). Set via modifier mesh in OrcaSlicer or PrusaSlicer.
Body 2 infill	12% Gyroid (mid zone 18-96mm)
Body 3 infill	8% Gyroid (TE zone 96-116.4mm)
Body 1 perimeters	4 perimeters = 1.60mm wall
Body 2 perimeters	3 perimeters = 1.20mm wall
Body 3 perimeters	2 perimeters = 0.80mm wall
Estimated time per rib	8 to 12 minutes
Total print time (110 ribs)	Approximately 15 to 22 hours
Effective density	Approximately 0.55 g/cm³ (44% of standard PLA)
Expected mass per rib	0.80 to 1.15g (adjust extrusion multiplier to achieve)

Print one test rib first. Verify mass is 0.80-1.15g. Verify LE bore accepts 2.0mm spar rod freely. Verify 2.2mm rod does NOT pass through bore. Do not batch-print 110 ribs until test rib passes all checks.

CHAPTER 5 — BILL OF MATERIALS — COMPLETE WING SYSTEM (ALL 8 WINGS)

The following covers all materials required to complete all eight wings including ribs, spars, membrane, covert flaps, VGs, fences, end plates, LE fairing strips, TE strips, and root attachment plates. Quantities include a 10% contingency allowance.

5.1 Printed Parts

Item	Quantity	Material	Notes
Universal wing rib — 120mm	110 ribs	LW-PLA (ColorFabb or equivalent)	~0.9g each = ~99g total

5.2 Carbon Fibre Structural

Item	Specification	Quantity	Notes
LE spar — W1,W2,W4 wings (6 wings)	2.0mm OD / 1.5mm ID pultruded CF tube	6 × 600mm	3.6m total + spare = order 4.5m
LE spar — W3 wings (2 wings)	2.5mm OD / 1.5mm ID pultruded CF tube	2 × 630mm	1.26m + spare = order 1.8m
Secondary spar — all 8 wings	2.0mm OD / 1.0mm ID pultruded CF tube	8 × 600mm	4.8m + spare = order 6.0m
LE fairing strip — all 8 wings	0.3mm × 6mm CFRP flat strip	8 × 600mm	4.8m + spare = order 6.0m
Trailing edge strip — all 8 wings	0.5mm × 4mm CFRP flat strip	8 × 600mm	4.8m + spare = order 6.0m
Wing fence material	0.2mm CFRP sheet	300mm × 200mm sheet	CNC or laser cut to shape
End plate material	0.5mm CFRP sheet	400mm × 300mm sheet	NACA 0008 profile, CNC cut

Item	Specification	Quantity	Notes
Root attachment plates	2.0mm CFRP sheet	8 plates 40mm × 30mm	CNC cut, M3 bolt holes

5.3 Membrane and Film Materials

Item	Specification	Quantity	Notes
Wing membrane — upper and lower	Ripstop nylon, 20 micron	2.2 m²	Grain chordwise. Buy 2.5m² to allow for cutting waste.
Film covert flaps — all rows	Ripstop nylon 20 micron OR Mylar/PET 25 micron	0.4 m²	Same material as membrane preferred. Cut into strips on assembly.

5.4 Fasteners and Adhesives

Item	Specification	Quantity	Notes
Thin CA adhesive (rib bonding)	Loctite 401 or equivalent	2 × 20ml bottles	Rib to spar bonding
Contact adhesive (membrane)	UHU Por or equivalent	1 × 100ml tin	Membrane to rib face bonding
Structural epoxy (root plates)	Araldite 2011 or Hysol EA9394	1 × 50ml syringe	Root plate to spar bonding
VG bonding film	3M 9460PC structural film adhesive	0.15 m²	Pre-cut to VG dimensions
M3 socket head cap screws	A4 stainless, 8mm length	32 screws	Root plate attachment, 4 per wing
M3 hex nuts	A4 stainless	32 nuts	Root plate attachment
M3 washers	A4 stainless	64 washers	Root plate attachment
Thread-locker	Loctite 243 medium strength	1 × 5ml bottle	All rotating fasteners

5.5 VG Array Components

Item	Specification	Quantity	Notes
VG vanes	0.3mm CFRP sheet, CNC laser cut	Per wing: approx 24 pairs	Counter-rotating delta pairs. Height 0.8% local chord (~1mm).
VG total all 8 wings	0.3mm CFRP sheet, 150mm × 150mm	1 sheet	Yields all VGs for all 8 wings

5.6 Mass Summary

Item	Mass	Notes
LW-PLA ribs (all 8 wings)	~99g	110 ribs × 0.9g average
Carbon spars (all 8 wings)	~22g	LE + secondary per wing
Membrane ripstop nylon	~24g	Upper + lower all 8 wings, 4g/m²
Film covert flaps	~4g	Three rows per wing, 20 micron film
LE fairing strips	~6g	0.3mm CFRP strip all 8 wings
TE strips	~8g	0.5mm CFRP strip all 8 wings
VG arrays	~5g	0.3mm CFRP all wings
Wing fences	~2g	0.2mm CFRP all wings
End plates	~10g	0.5mm CFRP NACA 0008 all wings
Root attachment plates	~8g	2mm CFRP 8 plates
Adhesives and fasteners	~8g	Estimated in-place mass
TOTAL 8-WING SYSTEM	~196g	Target 185-210g

CHAPTER 6 — ASSEMBLY INSTRUCTIONS — ONE COMPLETE WING

These instructions cover the assembly of one wing. Repeat for all eight. Left and right wings are mirror images — assemble the right side first, then mirror all jig settings for the left. Allow all adhesive bonds to cure fully before proceeding to the next step. Work on a flat, clean surface.

6.1 Tools Required

- Digital caliper (0.01mm resolution)
- 0.01g resolution jeweller's scale
- Flat assembly jig or flat board with dihedral angle blocks
- Impulse hammer or light pencil for natural frequency test
- Smartphone or oscilloscope for frequency measurement (apps: PhyPhox, Spectroid)
- Sharp scalpel or craft knife
- Scissors (for membrane cutting)
- Planform template cut from cardboard (trace from drawing for each wing)
- Steel rule and pencil
- Syringe applicators for CA and epoxy
- Activator spray for CA (optional but speeds cure)
- Masking tape (alignment aid)
- 220-grit sandpaper (edge finishing)

6.2 Preparation

STEP PREP-1 — Sort Ribs

Count out the required number of ribs for this wing (14 for W1, 13 for W2 and W4, 15 for W3).

Weigh each rib. Discard any outside the 0.80-1.15g range. Check LE bore with a 2.0mm rod — it must slide freely through. Check that a 2.2mm rod does NOT pass through.

Arrange in order from root to tip. Mark the root rib with a pencil dot.

STEP PREP-2 — Cut Spar to Length

Cut the LE spar tube to the wing span length: W1=550mm, W2=495mm, W3=578mm, W4=523mm.

Cut the secondary spar rod to span length minus 20mm (leaves 10mm clear each end).

Deburr both ends of all cut spars with 220-grit sandpaper wrapped around a finger.

Check both spars fit through all rib bores and notches smoothly.

STEP PREP-3 — Cut Planform Template

Cut a cardboard template of the wing planform from the wing drawing.

The template shows the exact shape the membrane will be trimmed to after bonding.

Label it with the wing number and LEFT or RIGHT.

Keep this template for the membrane trimming step.

6.3 Rib and Spar Assembly

STEP 1 — Thread Ribs onto LE Spar

Slide the root rib onto the LE spar first. The spar passes through the 2.10mm bore in the rib nose section. The rib should slide freely with no force required.

Slide all remaining ribs onto the spar in order, root to tip.

Space ribs at approximately 40mm centres in the root half of the span, 55mm centres in the tip half. Exact spacing is not critical — uniform distribution matters more than precise dimension.

Ensure the tip rib sits within 20mm of where the membrane tip edge will be trimmed. Check against the planform template.

Do not bond yet — check all ribs are oriented the same way (upper surface all same side).

STEP 2 — Set Dihedral and Bond Ribs

Place the assembly in the dihedral jig. Dihedral angles: W1=3°, W2=5°, W3=7°, W4=4°. A simple jig is a flat board with a wedge block under the tip end of the spar.

Apply thin CA to each rib-spar interface from both faces. Use a syringe applicator — capillary action draws the CA into the bore gap.

Hold each rib perpendicular to the spar for 10-15 seconds until CA grips.

Work root to tip. Allow 2 minutes full cure before moving to the next step.

Do not glue the root rib yet — it receives the root attachment plate in a later step.

STEP 3 — Install Secondary Spar

Drop the secondary spar rod into the semi-circular notches on the upper surface of all ribs. The rod should sit in the notches under gravity with zero force.

Apply thin CA to each notch contact point from above. Allow CA to wick into the notch around the rod.

The secondary spar should not be forced into any notch. If it does not drop in freely, lightly enlarge the notch with a 3mm drill or round file.

STEP 4 — Natural Frequency Test

THIS STEP IS MANDATORY BEFORE PROCEEDING.

Clamp the root end of the wing assembly in a vice or clamp (through the root attachment plate area). Wing should be horizontal, tip unsupported.

Open a frequency measurement app on a smartphone (PhyPhox Acoustic or Spectroid). Alternatively connect an accelerometer to the tip.

Tap the wingtip sharply with a fingertip. Observe the decay oscillation.

Read the frequency from the peak in the spectrum or from timing the decay cycles.

ACCEPT: 1.8 to 2.2 Hz.

If fn > 2.2 Hz (too stiff): replace LE spar with the next size down (W1/W2/W4: from 2.0mm OD try 1.5mm OD; W3: from 2.5mm OD try 2.0mm OD).

If fn < 1.8 Hz (too flexible): replace LE spar with next size up.

Record the measured fn for each wing. Wings at the same position should be within 0.1 Hz of each other.

6.4 VG Array Installation

STEP 5 — Attach VG Vanes

Install VGs before the membrane. Once the membrane is on, VG access is impossible.

VG position on the universal rib: X = 20.40mm from LE (centred in the VG recess).

Cut a strip of 3M 9460PC film adhesive to fit the VG recess footprint.

Bond VG vane pairs into the recess using the 9460PC film. Use the alignment dimple to centre the first pair.

VG height: 0.8% of local chord = approximately 1mm. VG cant angle: 18° to local flow direction. Counter-rotating pairs.

Press firmly for 30 seconds. Allow 1 hour before membrane application.

6.5 LE Fairing Strip

STEP 6 — Bond LE Fairing Strip

The 0.3mm × 6mm CFRP fairing strip bridges between rib noses along the leading edge.

Apply contact adhesive (UHU Por) to the upper surface of all rib nose sections and to one face of the fairing strip. Allow 60 seconds to become tacky.

Press the fairing strip against the leading edge of the rib array, running rib-to-rib across the full span. The strip sits on the upper surface of the rib noses and follows the spar.

Press firmly along the full length. Hold for 2 minutes.

Trim any excess at root and tip with a sharp scalpel.

This strip prevents membrane collapse between ribs at the leading edge and provides a continuous bonding surface for the upper membrane at the nose.

6.6 Membrane Application

STEP 7 — Apply Upper Membrane

Cut a membrane panel approximately 30mm larger than the wing planform in all directions. Grain direction: chordwise (parallel to the ribs).

Apply contact adhesive to the upper face of all ribs, the top of the LE fairing strip, and the top face of the TE strip (which will be bonded in the next step). Also apply adhesive to the corresponding areas of the membrane underside.

Allow 60-90 seconds. Both surfaces should be tacky but not wet.

Lay the membrane onto the wing assembly. Start at the leading edge — press the membrane firmly onto the LE fairing strip first, then work spanwise from root to tip, then chordwise to the TE.

CRITICAL: Do NOT stretch the membrane spanwise. Allow it to lie naturally — it should be just wrinkle-free, not taut.

Do NOT bond the inter-rib membrane spans to anything. Only the rib faces, LE strip, and TE strip receive adhesive. The membrane must be free to billow between ribs under aerodynamic pressure.

STEP 8 — Bond Trailing Edge Strip and Trim Membrane

Before bonding the TE strip, fold the membrane over the trailing edge of the last rib. Apply contact adhesive to the TE strip faces and the membrane in that zone.

Press the 0.5mm × 4mm CFRP TE strip onto the trailing edges of all ribs, sandwiching the membrane against them. Press firmly along the full span.

Using the planform template as a guide, trim the membrane to the wing planform shape with sharp scissors. Cut cleanly along the template edge.

Trim the leading edge membrane flush with the front of the LE fairing strip.

Trim the tip edge perpendicular to the span axis.

The wing planform is now defined by the trimmed membrane edge, not by the rib chord length. Ribs that extend past the membrane edge are fine — they are simply internal formers.

STEP 9 — Apply Lower Membrane

Repeat Steps 7 and 8 for the lower surface.

Bond the lower membrane to the lower faces of all ribs, the lower face of the TE strip, and the underside of the LE fairing strip at the leading edge.

The lower membrane should be bonded very slightly tighter than the upper — it should lie nearly flat when the wing is held horizontal.

Trim to the same planform template as the upper membrane.

CHECK: Hold the wing at the root and blow gently from the LE at approximately 1 m/s. The upper membrane should visibly billow 3-5mm above the rib profile. The lower membrane should remain nearly flat. If the upper does not billow: the tension is too high — carefully slit the bond at one or two inter-rib stations and re-bond with less tension.

6.7 Film Covert Flaps

STEP 10 — Cut and Bond Film Flap Strips

Cut three strips of 20-micron ripstop nylon (or Mylar) for each wing.

Strip dimensions: width = 9.60mm (flap chord), length = span coverage from 15% to 85% semi-span measured along the actual wing.

Apply contact adhesive to the FORWARD EDGE ONLY of each strip — the first 2mm of the strip width. Also apply to the flap ledge recesses on the rib upper surfaces.

Allow 60 seconds to become tacky.

Press the forward edge of each film strip into its corresponding ledge recess. Row 1 at 45mm, Row 2 at 63mm, Row 3 at 81mm from the LE.

The rear portion of each strip (7-8mm) must be COMPLETELY FREE. Do not apply any adhesive rearward of the ledge. Do not press the rear edge down.

CHECK: Lift each strip from its rear edge with a fingernail. It should hinge freely from the bond line. If it does not lift freely the rear is inadvertently bonded — carefully peel back and re-do.

The bond line IS the hinge. No piano wire, no spring, no magnet.

6.8 Wing Fences and End Plate

STEP 11 — Bond Wing Fences

Cut fence strips from 0.2mm CFRP sheet: height = 3% of local chord, chord extent = 10% to 72% of local chord.

Fence positions: W1 at 45% semi-span; W2 at 40%; W3 at 30% AND 60%; W4 at 50%.

Apply thin CA to the fence base and press onto the upper membrane surface perpendicular to the chord plane.

Verify perpendicularity within 2° using a small set square.

Hold for 30 seconds. Do not disturb for 5 minutes.

STEP 12 — Bond End Plate

The end plate is a NACA 0008 symmetric section, cut from 0.5mm CFRP.

Dimensions: approximately 15% of wing plan area. Height above chord plane: 8% semi-span. Height below: 8% semi-span.

The upper plate face is CANTED 15° outboard (not vertical). The lower plate is vertical.

Bond to the tip rib with CA and one M2 bolt through a pre-drilled hole.

Verify the 15° outboard cant on the upper plate with a protractor.

6.9 Root Attachment Plate

STEP 13 — Fit Root Attachment Plate

Bond the 2mm CFRP root attachment plate to the root rib and LE spar using Araldite 2011 structural epoxy.

The plate has a 2.10mm clearance hole for the LE spar and two M3 bolt holes for attachment to the pitch axis of the wing root mechanism.

Clamp for 60 minutes minimum cure. Do not load for 24 hours.

Bolt the root rib to the spar through the plate using M3 screws with thread-locker (Loctite 243).

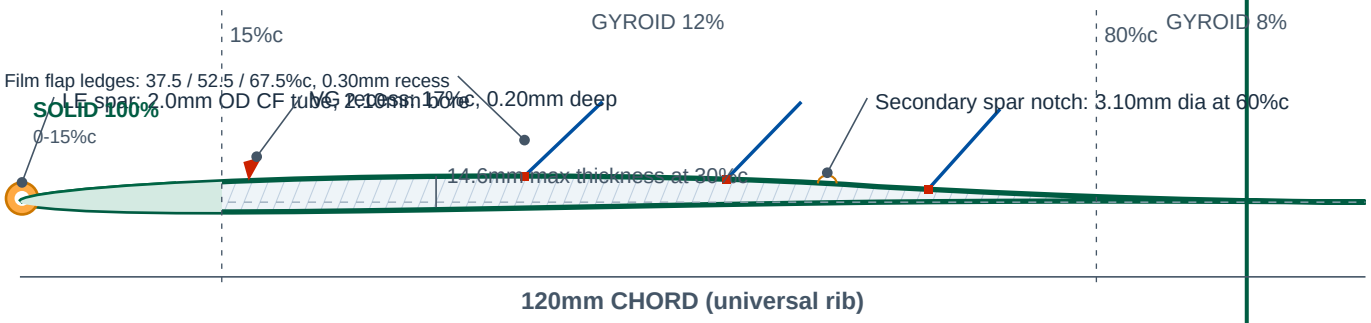
6.10 Final Quality Inspection

Parameter	Acceptance Criterion
Wing planform	Matches template within 2mm on all edges
Trailing edge	Sharp, less than 0.5mm radius, no step at membrane joint
Natural frequency	1.8-2.2 Hz measured and recorded

Parameter	Acceptance Criterion
LE bore	2.0mm spar slides freely; 2.2mm does not pass
Membrane billow	Upper surface billows 3-5mm at 1 m/s gentle airflow from LE
Film flap Row 1	Lifts freely when blown at approximately 15° AoA, 1 m/s
Film flap Row 2	Lifts freely at approximately 20° AoA
Film flap Row 3	Lifts freely at approximately 28° AoA
Film flap rear edges	Completely free throughout their length. No accidental bonding.
VG bonding	All VGs firmly bonded. No peeling at leading edge of any VG.
Wing fence	Perpendicular to chord plane within 2°
End plate cant	Upper plate 15° outboard within 2°
Root plate	Flat within 0.3mm. Bolts secure with thread-locker applied.
Wing mass	W1: 22-28g. W2: 20-26g. W3: 21-27g. W4: 18-24g.
Mirror symmetry	Left and right wings at same position within 2g mass

A wing that passes all quality checks is ready for installation into the wing root mechanism. A wing that fails any check should be repaired or remade before installation. The natural frequency check is the most important single criterion — it determines aerodynamic efficiency more than any other measurable parameter.

End of Wing Manufacturing Document. Blueprint schematic of S1223 rib cross-section follows.



KEY:

Solid LW-PLA nose zone (0-15%c) · Gyroid infill zones · Carbon fibre spar · LE fairing strip (0.30mm) · Film CF flap ledges (3 positions) · VG recess position · Adhesive bond line

MCO-1D — S1223 RIB CROSS-SECTION — UNIVERSAL 120mm DESIGN

LW-PLA gyroid 12% · Solid nose zone · Three film flap ledges · VG recess · LE and secondary spar bores

Schematic — not to scale